

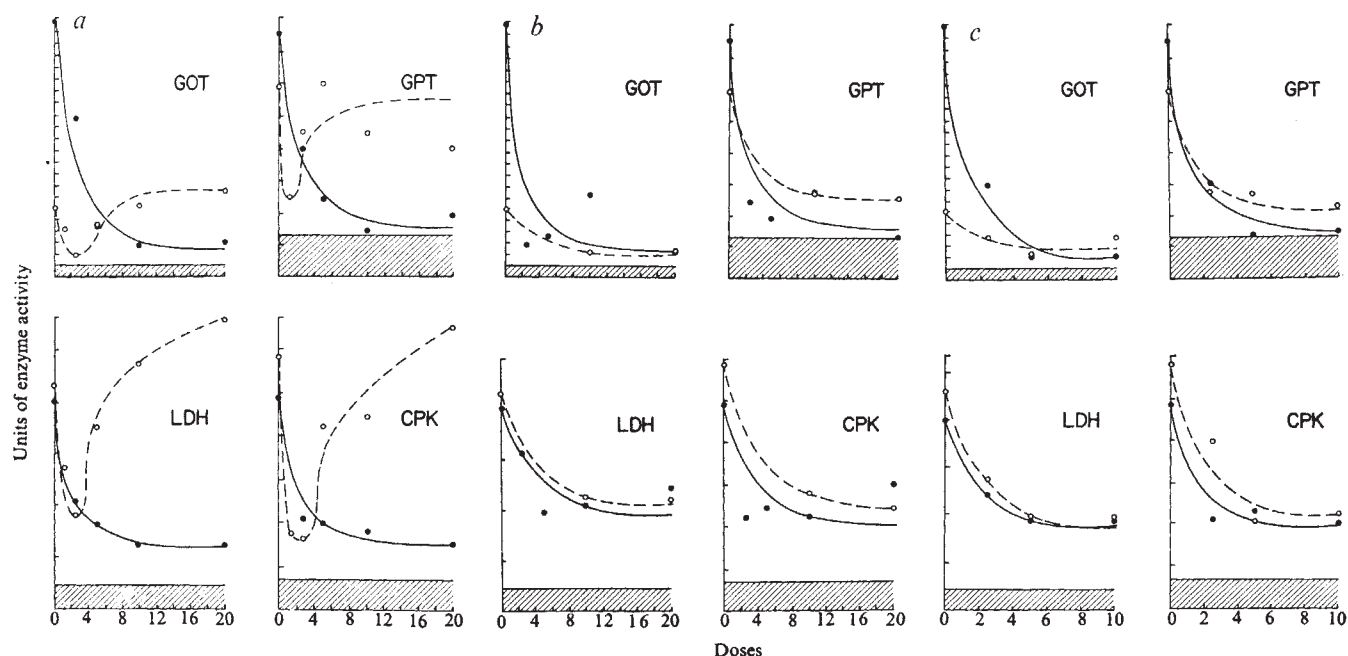


Fig. 1 Mean values (40–75 animals for each point) of plasma enzyme levels of GOT, GPT, LDH and CPK in aorta-ligated rats given a single dose of 5-HT (●) or noradrenaline (○), pretreated by: *a*, imipramine; *b*, phenoxybenzamine; or *c*, chlorpromazine. Doses of pretreatment drugs on abscissa are given in mg kg^{-1} . Divisions on the ordinate are GOT 100, GPT 20, LDH 100 and CPK 100 U of enzyme. Zero point on the abscissa indicates enzyme level in control animals ligated and given 5-HT or noradrenaline but no pretreatment drug; cross-hatched region indicates maximal upper range (on ordinate) of enzyme level in normal ligature-only rats, which is the same as in normal rats.

smooth muscle of the abnormal (underperfused) intramuscular arterial vessels of our model.

We conclude that this is a useful animal test system for evaluating interaction of drugs which affect muscle, presumably by acting on the intramuscular arterial system. If the muscle-damaging mechanism in our model is, in principle, analogous to that in Duchenne muscular dystrophy, these experiments provide screening data of possible clinical relevance. Because local metabolic factors are thought to be producing maximal vascular deconstriction in ischaemic regions of normal muscle⁷, blockers of amine-induced vasoconstriction in normal persons may only be able to open collateral feeding vessels. In Duchenne dystrophy muscle, however, we propose that there could be an inherited defect making small arterial vessels hyposensitive to deconstrictive local metabolic factors and/or overly sensitive to other constrictive ones such as vasoactive amines, defects for which amine-blockers might be at least partially corrective.

The pertinence of our model, involving aorta ligation and vasoactive amine in low dosage, to Duchenne dystrophy has recently been questioned and an alternative one proposed—IP plus 5-HT at very high doses (10 times that used in our experiments) in non-ligated rats¹¹. Pretreatment with IP had only a worsening effect (histologically) on muscle damage produced by the very high doses of 5-HT¹¹. Nevertheless, in Duchenne dystrophy patients we favour the mechanism/model of a small amount of vasoactive substance interacting with an underlying arterial vessel abnormality because they have no systemic evidence of large circulating amounts of vasoactive amine.

W. KING ENGEL
E. CAROLYN DERRER

Medical Neurology Branch,
National Institute of Neurological Diseases and Stroke,
National Institutes of Health,
Bethesda, Maryland 20014

Received August 14, 1974; revised January 6, 1975.

¹ Engel, W. K., in *Neurohistochemistry* (edit. by Adams, C.), 622–672 (Elsevier, Amsterdam, 1965); *Pediatr. Clin. North Am.*, **14**, 963–995 (1967); *Rev. Neurol. Paris*, **124**, 291–298 (1971); in *The Striated Muscle*, 453–472 (Williams and Wilkins, Baltimore, 1973).

² Hathaway, P. W., Engel, W. K., and Zellweger, H., *Archs Neurol.* **22**, 365–376 (1970).

- ³ Mendell, J. R., Engel, W. K., and Derrer, E. C., *Science*, **172**, 1143–1145 (1971).
- ⁴ Mendell, J. R., Engel, W. K., and Derrer, E. C., *Nature*, **239**, 522–524 (1972).
- ⁵ Quastel, D. M. J., and Hackett, J. T., in *The Structure and Function of Muscle*, IV (edit. by Bourne, G. H.), 2–153 (Academic Press, New York, 1973).
- ⁶ Goodman, L. S., and Gilman, A., in *The Pharmacological Basis of Therapeutics*, fourth ed. (Macmillan, New York, 1970).
- ⁷ Sourkes, T. L., in *Basic Neurochemistry* (edit. by Albers, R. W., Siegel, G. J., Katzman, R., and Agranoff, B. W.), ch. 29, 581 (Little, Brown, Boston, 1972).
- ⁸ Axelrod, J., *Seminars in Psychol.*, **4**, 199–210 (1972).
- ⁹ Whelan, R. F., in *Control of the Peripheral Circulation in Man* (Thomas, Springfield, Illinois, 1967).
- ¹⁰ Chase, T. N., and Murphy, D. L., in *Ann. Rev. Pharmac.* (edit. by Elliott, H. W., Okun, R., and George, R.), **13**, 181–197 (Annual Reviews, Inc., Palo Alto, California, 1973).
- ¹¹ Parker, J. M., and Mendell, J. R., *Nature*, **247**, 103–104 (1974).

Fluorescence techniques for following interactions of microtubule subunits and membranes

THERE is increasing evidence that several cell functions are controlled by the state of polymerisation of microtubules (MTs) and by the interaction of MTs with membranes. For example, previous studies in our laboratory have shown that colchicine-sensitive structures, presumably MTs, determine the topographical organisation of cell membrane components^{1–3}. Conditions for the polymerisation of MTs *in vitro* have been described⁴. The MT subunit has been identified by centrifugation studies as a 6S dimer of approximately 110,000 daltons⁵. An additional 36S component is required for MT assembly⁶. The established procedures used, however, to record MT polymerisation (light scattering and viscosity measurements supplemented by negative staining and electron microscopy) do not reveal details of the interactions between subunits undergoing polymerisation and cannot be applied in the presence of elements such as membranes that contribute separately to light scatter and viscosity.

We report here a new method to monitor MT polymerisation and to investigate interactions between MTs and membranes. Our approach involves long range resonance energy transfer between different fluorescent moieties separately conjugated to different MT subunits or to membranes. The method permits study of weak or readily dissociable interactions between membranes and MTs or between subunits, which may not

survive techniques that physically separate or dilute the complexes.

Resonance energy transfer occurs when a chromophore which is excited by the absorption of energy transmits the energy to another (acceptor) chromophore some distance away. This transmission requires the overlap of the emission spectrum of the first (donor) chromophore with the absorption spectrum of the acceptor, but it does not involve the actual reabsorption of light by the acceptor. It is therefore referred to as non-radiative or resonance energy transfer. Resonance transfer requires that the distance between the chromophores be relatively close (not exceeding 100 Å) and in fact quantitative theory allows one to estimate the distance between chromophores from the extent of energy transfer⁷⁻¹⁰.

In our experiments fluorescein isothiocyanate (FITC) was used to label one population of MT subunits (protein dimers called tubulin and designated T) or membranes, and rhodamine isothiocyanate (RITC) was used to label a second population of tubulin. These chromophores were chosen because they bind covalently and because the emission spectrum of fluorescein (donor) extensively overlaps the absorption spectrum of rhodamine (acceptor). In the non-polymerised state a mixture of fluorescein-labelled (F-T) and rhodamine-labelled (R-T) tubulins does not show resonance transfer. With polymerisation (or aggregation) of MT-subunits, however, the chromophores are brought sufficiently close together so that energy transfer occurs. The degree of transfer is thus dependent on the number of units polymerised. Figure 1 shows a time course of the polymerisation of MTs followed by light scattering and resonance energy transfer. The progressive increase in scatter was the same whether or not dye was conjugated with the tubulin, indicating that these low concentrations of bound chromophores do not interfere with MT assembly. At the same time resonance transfer of energy from fluorescein to rhodamine was measured by exciting the sample at 480 nm, and recording the emission spectrum between 500 and 620 nm (to include

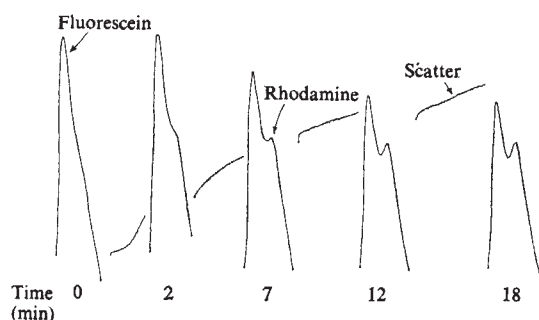


Fig. 1 Time course of light scattering and fluorescence emission spectra during MT polymerisation. Tubulin (T) was purified from rabbit brain by successive cycles of polymerisation and depolymerisation as described by Shalanski¹¹. Protein-dye conjugates were formed by incubation of purified protein (4 mg ml⁻¹) with fluorescein isothiocyanate (FITC) or rhodamine isothiocyanate (RITC) for 30 min at 4 °C. Incubation was stopped by separating the conjugates from the free dye by column chromatography (Sephadex G-25). Dye to protein molar ratios (D/P) were determined by absorbance measurements using extinction coefficients of the bound dyes¹² and protein measurements by the method of Lowry¹³. The absorbances of the mixtures followed spectroscopically did not exceed 0.20 absorbance units. In this experiment incubations of 10 µg dye per mg protein and 5 µg mg⁻¹ protein for RITC and FITC respectively gave approximate D/P values of 0.35 for R-T and 0.08 for F-T. Tyndall scatter was measured at 650 nm using parallel polarised light, monitored at 90° to the incident beam. (This assures a high signal to noise ratio.) Viscosity changes (not shown) agreed closely with the time course of scatter measurements. At intervals the latter were interrupted and emission spectra obtained by exciting the sample at 480 nm and scanning from 500 to 650 nm. Corrected spectra were obtained using a Perkin-Elmer MPF4 fluorometer. At zero time a single peak representing only fluorescein emission was obtained. A rhodamine emission peak developed progressively as resonance energy transfer occurred between the polymerising fluorescein and rhodamine-labelled subunits. See text.

both fluorescein and rhodamine emissions). The sample was excited at 480 nm instead of at the excitation maximum of 496 nm to minimise the direct excitation of rhodamine. Before polymerisation (Fig. 1, zero time) a simple fluorescein emission spectrum was observed. There was no energy transfer and thus no rhodamine emission. Later (Fig. 1, 2 min and later) as subunits separately labelled with F and R were brought together, resonance transfer occurred and the characteristic emission peak of rhodamine progressively increased. Simultaneously, since resonance transfer and emission compete for the excited state, a decline in fluorescein emission was observed. When F-T alone (no R-T present) were polymerised, no decrease in fluorescein emission occurred. The depopulation of the fluorescein excited state by energy transfer to rhodamine is also reflected in a decrease in the excited state lifetime. Fluorescent lifetime measurements¹⁴ showed a decrease from 5.15 ns for MT assembled from F-T dimers alone to 3.64 ns for maximally (> 70%) polymerised MTs assembled from the F-T plus R-T mixture. From these data a mean distance between chromophores of 44 Å was calculated⁷. The same distance was obtained by calculation from the energy transfer spectrum of maximally polymerised MT obtained in Fig. 1, and is consistent with the known geometry of MTs (Fig. 2).

Energy transfer did not occur when depolymerised F- and R-labelled subunits were incubated under conditions in which polymerisation did not occur (at 4 °C, or in the presence of 4 mM calcium chloride, or in the absence of GTP). But we observed interactions between subunits in the presence of colchicine, a plant alkaloid that binds with tubulin subunits and is thought to prevent their association into microtubules. A tenfold molar excess of colchicine slowed markedly the increase in the light scattering. None the less, in the presence of colchicine resonance transfer increased progressively with time and to nearly the same degree as in the absence of the alkaloid. This indicates that most of the MT subunits are probably associated into small aggregates. These aggregates seem to be readily dissociable: the addition of unlabelled tubulin to the colchicine mixture, or simple dilution, led to a rapid fall in energy transfer. In contrast, neither unlabelled tubulin nor dilution altered the energy transfer of polymerised MT. Thus, the effect of colchicine on the interaction of subunits was different from the effects of cold or calcium which prevent polymerisation and the development of energy transfer and light scattering.

Preliminary experiments have also demonstrated interactions of MT and/or tubulin with isolated cytoplasmic membranes. Cytoplasmic membranes were isolated from rabbit polymorphonuclear leukocytes as previously described⁹ and incubated with FITC at approximately 25 µg FITC per 100 µg membrane protein for 10 min at 4 °C. The labelled membranes were then washed free of unreacted FITC by recentrifugation through a discontinuous sucrose gradient. R-T was prepared as before and mixed with the labelled membrane. At 4 °C virtually no energy transfer took place between F-labelled membranes and R-labelled tubulin. At 37 °C polymerisation of R-T occurred, and resonance energy transfer between F-membranes and R-tubulin developed in parallel with the increase in light scatter. This transfer was not diminished by the addition of unlabelled tubulin subsequent to polymerisation. Similar experiments using rhodamine-labelled bovine serum albumin and fluorescein-labelled membranes did not reveal energy transfer at any temperature. These experiments suggest a specific association of polymerised MT with membrane. Energy transfer between R-T and F-membranes increased progressively with time in the presence of colchicine, just as it did between R-T and F-T with colchicine. Further, addition of unlabelled tubulin to the membrane-tubulin complex formed in the presence of colchicine (10⁻⁴ M) rapidly reduced the resonance transfer. Thus, in the presence of colchicine the association of tubulin with membranes, like the association of tubulin with tubulin seems readily reversible, and is distinct from the association formed in the absence of the alkaloid. The high

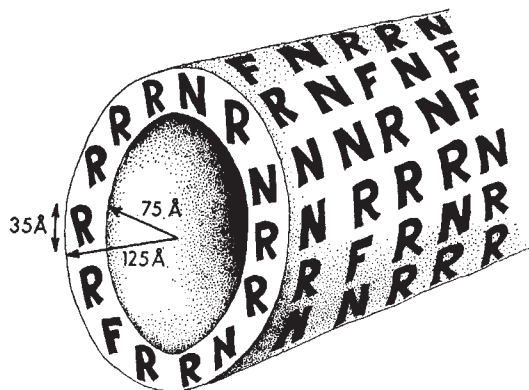


Fig. 2 Schematic diagram of microtubule and disposition of constituent fluorescent labelled subunits. The diagrammatic tubule was generated using the known geometry of MTs¹⁵ and distributing tubulin dimers labelled with fluorescein (F), rhodamine (R) and unlabelled dimers (N) in the tubule according to a list of random numbers. The proportions of F, R and N were calculated from the D/P ratios of F-T (0.15) and R-T (0.46) measured in a maximally (> 70%) polymerised suspension of MT, assuming that the association of dye molecules with tubulin follows a Poisson distribution¹⁶. In this mixture, the distance between F and R calculated from resonance transfer as described in the text is approximately 40 Å. This is consistent with the dimensions of the MT as shown, which is made up of protein dimers with a cross-sectional diameter of roughly 40 Å. This correspondence suggests that resonance transfer occurs only between immediately adjacent subunits.

efficiency of energy transfer observed (> 50%) indicates considerable interaction of the tubulin and membrane, and suggests that some labelled constituent of the membrane preparation could serve as a nucleation centre for MT polymerisation.

In summary the use of long range non-radiative energy transfer provides a new approach to the study of MY polymerisation. This method gives information on the geometry and interactions of subunits and of interactions between MTs and other cell constituents. It has been applied to demonstrate loosely associated tubulin aggregates formed in the presence of colchicine, and binding of microtubules to isolated membranes, neither of which has been observed with conventional techniques. These experiments are now being extended to explore the specificity and number of membrane binding sites for tubulin under different physiological conditions. The approach should prove useful in the study of other polymerisations (for example, microfilaments), particularly of their interactions in mixed systems containing membranes.

We thank Dr S. Fernandez for assistance in fluorescent lifetime measurements. This work was supported by a grant from the National Institutes of Health.

JOEL S. BECKER
JANET M. OLIVER
RICHARD D. BERLIN

Department of Physiology,
Schools of Medicine and Dental Medicine,
University of Connecticut,
Farmington, Connecticut 06032

Received November 22, 1974; revised February 3, 1975.

- ¹ Berlin, R. D., Oliver, J. M., Ukena, T. E., and Yin, H. H., *Nature*, **247**, 45-46 (1974).
- ² Oliver, J. M., Ukena, T. E., and Berlin, R. D., *Proc. natn. Acad. Sci. U.S.A.*, **71**, 394-398 (1974).
- ³ Ukena, T. E., Borysenko, J. Z., Karnovsky, M. J., and Berlin, R. D., *J. Cell Biol.*, **61**, 70-82 (1974).
- ⁴ Weisenberg, R. C., *Science*, **177**, 1196 (1972).
- ⁵ Weisenberg, R. C., Borisy, G. G., and Taylor, E. W., *Biochemistry*, **12**, 4466-4479 (1968).
- ⁶ Kirschner, M. W., Williams, R. C., Weingarten, M., and Gerhart, J. C., *Proc. natn. Acad. Sci. U.S.A.*, **71**, 1159-1163 (1974).
- ⁷ Forster, T. H., *Disc. Faraday Soc.*, **27**, 7-17 (1959).
- ⁸ Wu, C.-W., and Stryer, L., *Proc. natn. Acad. Sci. U.S.A.*, **69**, 1104-1108 (1972).
- ⁹ Bunting, J. R., and Cathou, R. E., *J. molec. Biol.*, **77**, 223-235 (1973).

- ¹⁰ Gennis, L. S., Gennis, R. B., and Cantor, C. R., *Biochemistry*, **11**, 2517-2524 (1972).
- ¹¹ Shelanski, M. L., Gaskin, F., and Cantor, C. R., *Proc. natn. Acad. Sci. U.S.A.*, **70**, 765-768 (1973).
- ¹² Chen, R. F., *Archs Biochem. Biophys.*, **133**, 263-276 (1969).
- ¹³ Lowry, O. H., Rosebrough, N. J., Farr, A. L., and Randall, R. J., *J. biol. Chem.*, **193**, 265-275 (1951).
- ¹⁴ Yguerabide, J., *Methods in enzymology*, **26**, Part C, 498-578, (Academic, New York, 1972).
- ¹⁵ Olmsted, J. B., and Borisy, G. G., *Ann. Rev. Biochem.*, **42**, 507-540 (1973).
- ¹⁶ Gennis, R. B., and Cantor, C. R., *Biochemistry*, **11**, 2509-2517 (1972).

Lateral compressibility and penetration into phospholipid monolayers and bilayer membranes

THE phospholipid bilayer¹ in certain biological membranes is maintained so that the range of temperature over which the gel-to-liquid crystal transition occurs includes the environmental temperature². As a result, clusters of phospholipid molecules in crystalline and liquid-crystalline states¹ coexist in the membrane and the isothermal lateral compressibility of the membrane lipids is enhanced³. Any increase in compressibility should facilitate insertion of foreign molecules into the bilayer thereby affecting transport across the membrane. Indeed, there is evidence that transport of ions^{4,5} and sugars³, and penetration of an enzyme⁶ is increased when the chain-melting transition of the lipids occurs. The lateral compressibility of phospholipid bilayers has not been measured, however, and there is no direct evidence for an increase in compressibility at the point where the gel-to-liquid crystal phase transition occurs. In order to measure lateral compressibility, compression solely in the plane of the bilayer is required. Such directed compression of bilayers will be difficult but it can be readily achieved with monolayers at the air-water interface. Since the molecular packing in such monolayers of phospholipids is equivalent to that in bilayers dispersed in excess water^{1,7}, this model system is particularly convenient for exploring the role of lateral compressibility. Here we show (1) how the compressibility of lecithin monolayers varies with packing density and changes at the chain-melting transition, and (2) how penetration of a hydrophobic protein is dependent on the compressibility.

The procedures used to obtain the surface pressure (π)-molecular area (A) isotherms for chromatographically pure 1,2-dipalmitoyl phosphatidylcholine (lecithin) have been described before^{7,8}. The preparation and adsorption characteristics of 1-¹⁴C-acetyl- β -casein A have also been published^{9,10}: this protein is very hydrophobic and its behaviour at the air-water interface is like that of proteins which are associated with lipids *in vivo* and quite different from that of highly water-soluble globular proteins. The penetration experiments were performed at constant surface area by maintaining the lecithin monolayer at an initial A with a barrier and injecting the protein into the substrate. The initial substrate concentration of protein was 1×10^{-4} g per 100 ml and the changes in π and surface concentration of protein (Γ) were monitored during adsorption with a Wilhelmy plate and gas-flow counter¹¹, respectively.

Figure 1a shows the π - A curves for dipalmitoyl lecithin spread at 20 °C and 30 °C on to a phosphate buffer ($I = 0.1$, pH 7); the isotherms are similar to those for a 0.1 M NaCl substrate⁷. The inflection at $\pi = 5.6$ mN m⁻¹ and $A = 77$ Å² per molecule (this point is difficult to reproduce and is sensitive to the rate of compression) is a result of the onset of two-dimensional crystallisation^{1,7} and over the region where the π - A curve is more-or-less horizontal the monolayer consists of a mixture of crystalline and liquid-crystalline domains. Because the transition is equivalent to that occurring in bilayers, the critical temperature^{1,7} for the monolayer transition is the same as the bilayer transition temperature (41 °C). The monolayer phase transition is probably best described as "diffuse" first order¹²⁻¹⁶ because the compressibility ($C = -1/A(\delta A/\delta \pi)_T$) plotted as a function of A (Fig. 1b) does not become infinite at the onset of condensation¹⁷. The form of the plot of C against A is the